# Forces & Equilibrium

## Section 1: Resolving Forces in Two Dimensions

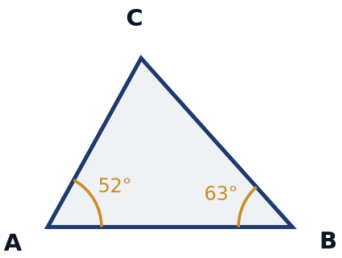
**1** A force of magnitude 50 N acts at an angle of $30^\circ$ above the horizontal. Find its horizontal and vertical components.

**a.** Horizontal: $43.3$ N, Vertical: $25.0$ N

**b.** Horizontal: $25.0$ N, Vertical: $43.3$ N

**c.** Horizontal: $35.4$ N, Vertical: $35.4$ N

**d.** Horizontal: $50.0$ N, Vertical: $0.0$ N




**2** Two forces $F_1 = [8,\ -3]$ N and $F_2 = [-5,\ 7]$ N act on a particle. What is the magnitude of the resultant force?

**a.** $5$ N

**b.** $7$ N

**c.** $\sqrt{41}$ N

**d.** $12$ N

**3** A particle is held in equilibrium by three coplanar forces: $P$ acting East, $Q$ acting North, and a force of 13 N acting at an angle $\theta$ South of West where $\tan\theta = 5/12$. Find the magnitude of $P$.

**a.** $12$ N

**b.** $5$ N

**c.** $13$ N

**d.** $8$ N

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## Section 2: Friction & Limiting Equilibrium

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**4** A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10\ \text{m/s}^2$, what is the maximum frictional force that can act on the block?

**a.** 21 N

**b.** 60 N

**c.** 18 N

**d.** 24 N

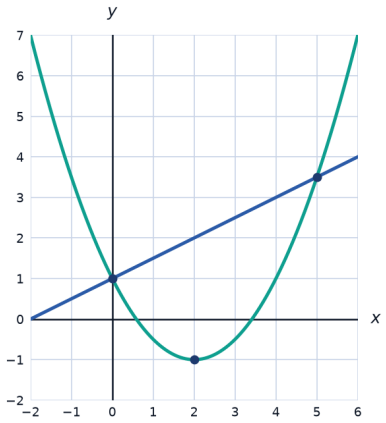
**5** A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\max} = 21$ N). What is the actual frictional force acting on the block?

**a.** 15 N

**b.** 21 N

**c.** 0 N

**d.** 6 N




**6** A particle of mass 4 kg is placed on a rough plane inclined at $30^\circ$ to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction $\mu$.

**a.** $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$

**b.** $\mu = \sin(30^\circ) = 0.5$

**c.** $\mu = \cos(30^\circ) \approx 0.866$

**d.** $\mu = 0.25$

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## Section 3: Inclined Planes & Equilibrium

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**7** A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at $30^\circ$ to the horizontal by a force $P$ acting parallel to the slope. Taking $g = 10\ \text{m/s}^2$, find $P$.

**a.** 50 N

**b.** 86.6 N

**c.** 100 N

**d.** 25 N

**8** For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?

**a.** 86.6 N ($50\sqrt{3}$ N)

**b.** 50 N

**c.** 100 N

**d.** 75 N

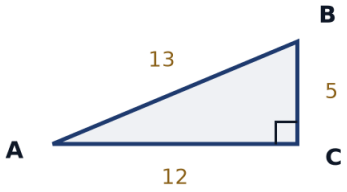
**9** A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10\ \text{m/s}^2$, find the angle of inclination $\alpha$ of the plane.

**a.** $\alpha = \arccos(0.8) \approx 36.9^\circ$

**b.** $\alpha = 30^\circ$

**c.** $\alpha = 45^\circ$

**d.** $\alpha = 60^\circ$




## Section 4: Lami's Theorem & Connected Systems

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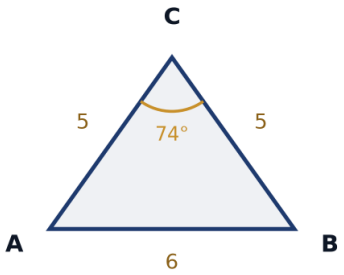
**10** A lamp of weight $W = 60$ N is suspended by two light strings making angles of $30^\circ$ and $60^\circ$ with the horizontal ceiling. Find the tension in the steeper string.

**a.** $30\sqrt{3}$ N $\approx 52.0$ N

**b.** 30 N

**c.** 60 N

**d.** 45 N




**11** A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s$^2$)

**a.** 20 N

**b.** 50 N

**c.** 30 N

**d.** 10 N

**12** A ring of mass $m$ is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension $T$ at an angle of $45^\circ$ above the wire. Find the value of $T$ (in terms of $mg$) when the ring is in limiting equilibrium.

**a.** $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$

**b.** $\dfrac{\sqrt{2}}{2}mg$

**c.** $0.4mg$

**d.** $mg$

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